



MALAYSIAN INSTITUTE OF INFORMATION TECHNOLOGY

MARCH 2026 SEMESTER

CLOUD COMPUTING (IIB43203)

GROUP PROJECT ASSIGNMENT

PREPARED BY:

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SUBMISSION DATE:

Sunday, 7 June 2026

Group Project Assignment

For this group project, you need to form a group of a maximum of four members and a minimum of two members

Deliverables must be:

1. Write a Report and a step-by-step guide for the reader to follow and replicate your project.
2. Create a video presentation about your project (All members must present) and publish the video presentation to your YouTube channel (put the link to the video in your report)
3. Save all your project materials, including the report, into your GitHub account.

Your task for this group project:

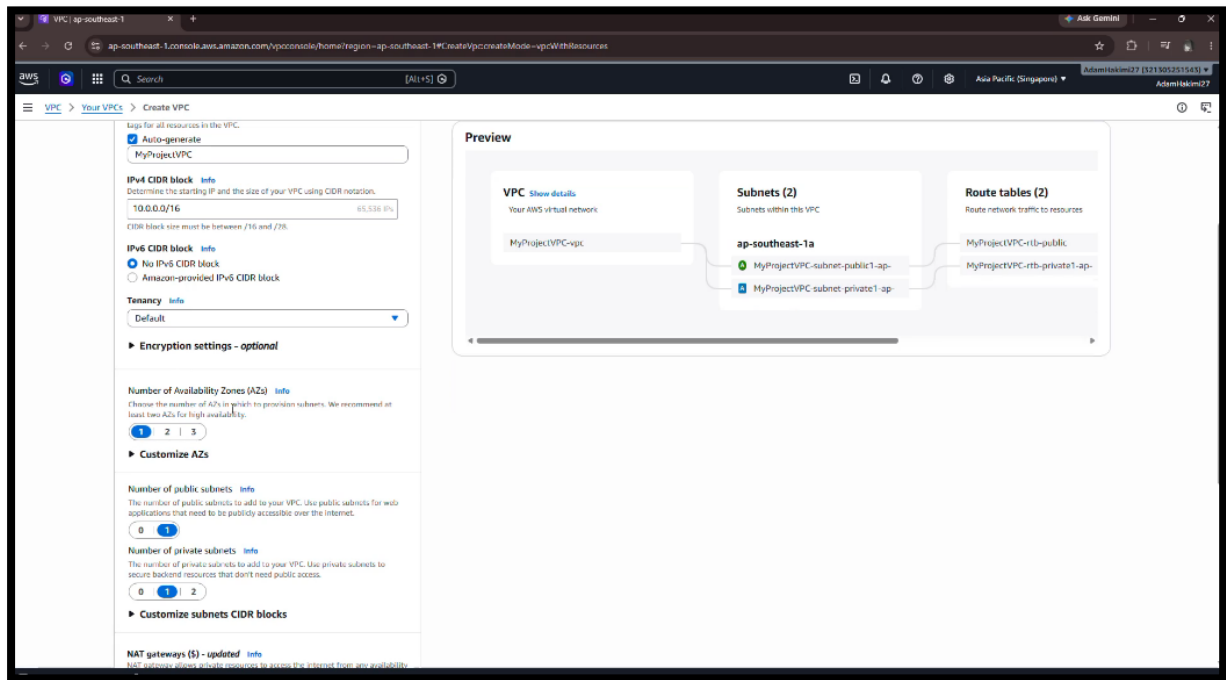
Using your AWS trial account, create a web application or a website based on Compute, Storage, Networking, and Database AWS Services

Marking criteria:

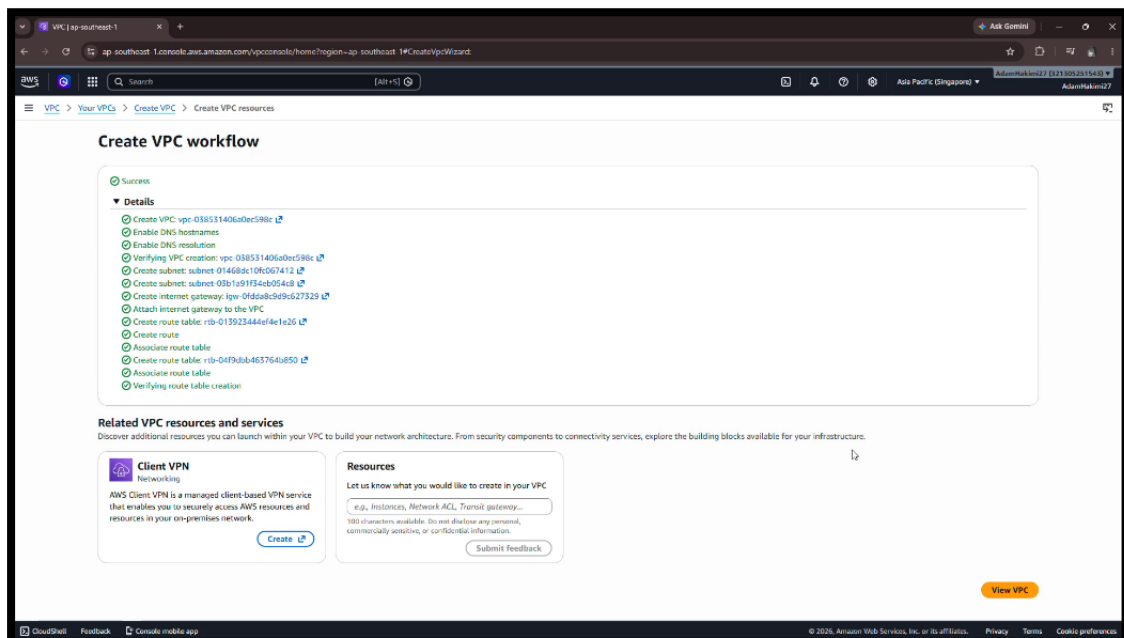
- Comprehensive instructional/tutorial guide and project reporting
- Comprehensive video tutorial explanation (ALL Team Members Must Present)
- Good quality presentation materials
- Good Creativity
- Show the outcome based on teamwork

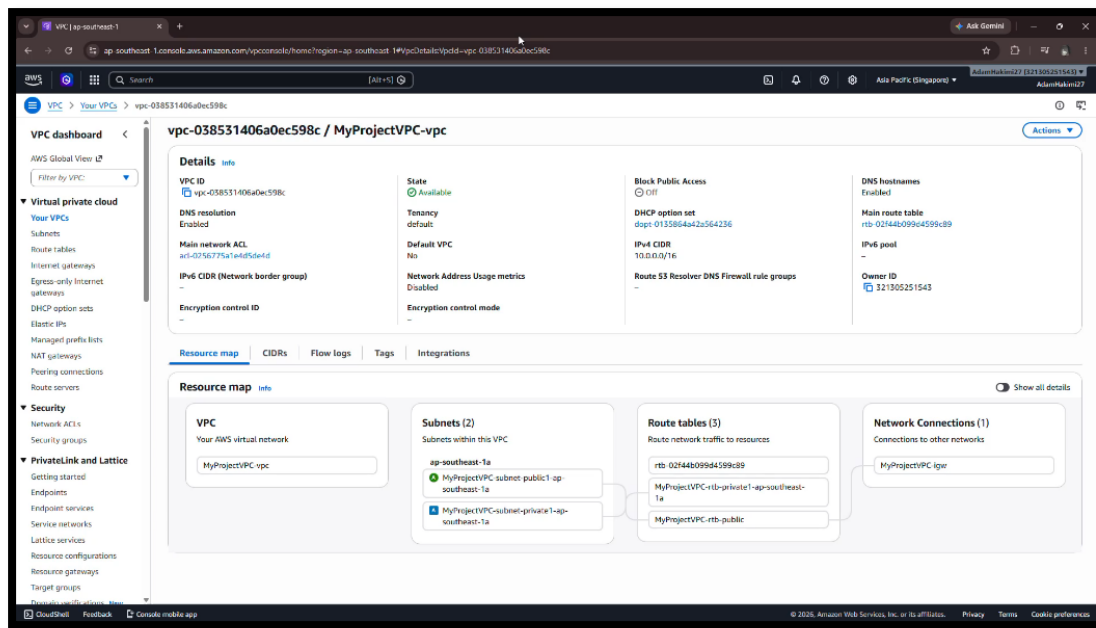
Please start early, and all the best!

Creating VPC



Creating VPC. Renaming the VPC to MyProjectVPC. Keep the IPv4 CIDR block as default. Change the Number of Availability zones to 1 and all the rest will follow to 1 like the number public of subnets and number of private subnets.





This is after you have done creating the VPC.

Creating EC2 Security Group

The screenshot shows the 'Create security group' page in the AWS Management Console. The page title is 'Create security group' with an 'info' link. Below the title is a descriptive sentence: 'A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.' The form is divided into a 'Basic details' section. It contains three input fields: 'Security group name' with the value 'WebServerSG', 'Description' with the value 'Security Group For Website Serve', and 'VPC' with a dropdown menu showing 'vpc-08f9af1de1a7de2a1'. There are 'info' links next to each field label. A note below the name field states 'Name cannot be edited after creation.'

You can name the security group name based on you project to make a different from other security group name.

The screenshot shows the 'Inbound rules' configuration page. It features a table with columns: 'Type', 'Protocol', 'Port range', 'Source', and 'Description - optional'. There are three rows of rules configured: 1) SSH (Type), TCP (Protocol), Port 22 (Port range), Source 'Anywhere...' (Source), and '0.0.0.0/0' (Source IP). 2) HTTP (Type), TCP (Protocol), Port 80 (Port range), Source 'Anywhere...', and '0.0.0.0/0'. 3) HTTPS (Type), TCP (Protocol), Port 443 (Port range), Source 'Anywhere...', and '0.0.0.0/0'. Each row has a 'Delete' button. An 'Add rule' button is at the bottom left. A warning banner at the bottom states: 'Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.'

Three inbound rules were configured in the Security Group to support the functionality of the web server. The SSH rule (Port 22) allows administrators to remotely access the EC2 instance for configuration and maintenance purposes. The HTTP rule (Port 80) enables users to access the website through a web browser. The HTTPS rules (Port 443) provides secure encrypted communication between the website and users, ensuring data confidentiality and security. Together, these rules allow server management and web access while supporting secure communication.

EC2 Launch Instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name
WebsiteServer

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Search our full catalog including 1000s of application and OS images

RecentsQuick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0543c0d4e114be7 (64-bit x86), uefi-optional / ami-04ff914164018330c (64-bit ARM), uefi
Virtualization: hvm | TPM: enabled: true | Root device type: xfs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260526.0 x86_64 HVM kernel-6.1

Architecture
64-bit (x86)

Boot mode
uefi-optional

AMI ID
ami-0543c0d4e114be7

Publish Date
2026-05-21

Username
ec2-user

Verified provider

Summary

Number of instances
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.11...read more
ami-0543c0d4e114be7

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

CancelLaunch instancePreview code

Instance type

InfoGet advice

Instance type

t3.micro

Free tier eligible

Family: t3 | 2 vCPU | 1 GiB Memory | Current generation: true | On-Demand Windows base pricing: 0.0224 USD per Hour | On-Demand SUSE base pricing: 0.0132 USD per Hour | On-Demand Linux base pricing: 0.0132 USD per Hour | On-Demand Ubuntu Pro base pricing: 0.0167 USD per Hour | On-Demand RHEL base pricing: 0.042 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

projectwebsite-key

Create new key pair

Network settings

Info

Network

Info

vpc-08f9ef1de1e7de2a1

Subnet

Info

No preference (Default: subnet in any availability zone)

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-3' with the following rules:

☒ Allow SSH traffic from

Help you connect to your instance

Anywhere
0.0.0.0/0

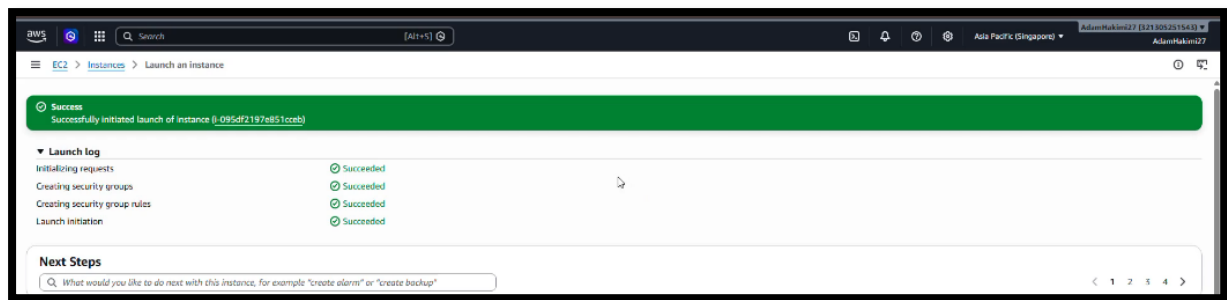
☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.



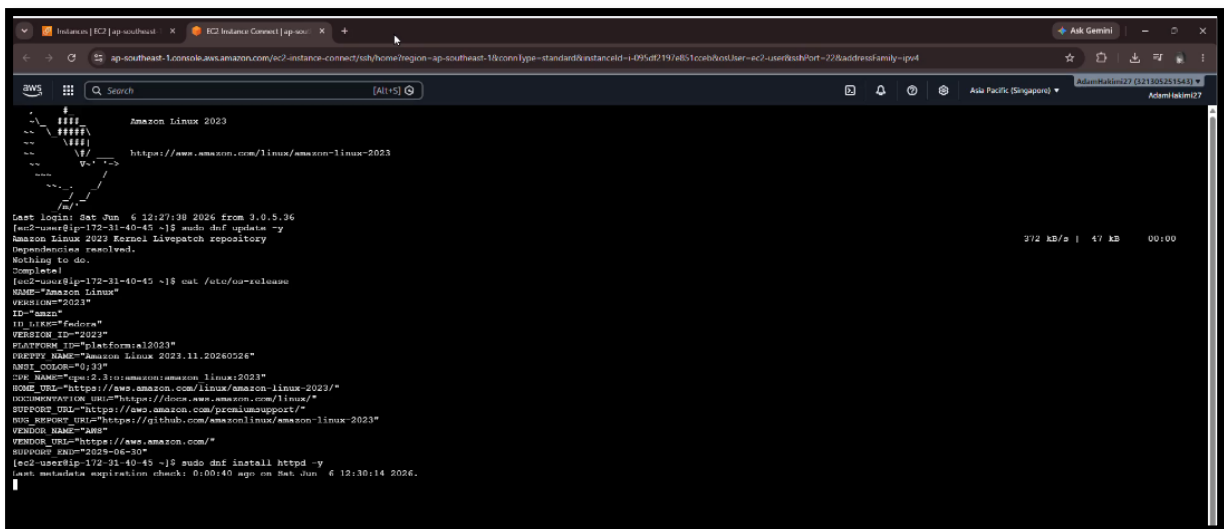
Connect to EC2 Instance

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\anaqi> ssh -i projectwebsite-key.pem ec2-user@172.31.40.45
Warning: Identity file projectwebsite-key.pem not accessible: No such file or directory.
PS C:\Users\anaqi> ssh -i projectwebsite-key.pem ec2-user@18.143.107.100
Warning: Identity file projectwebsite-key.pem not accessible: No such file or directory.
The authenticity of host '18.143.107.100 (18.143.107.100)' can't be established.
ED25519 key fingerprint is SHA256:bvIZR7dyOjH9ls+D5xRFry0QqLJHPiLCm0t6NRQnPL4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.143.107.100' (ED25519) to the list of known hosts.
ec2-user@18.143.107.100: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
PS C:\Users\anaqi> |
```

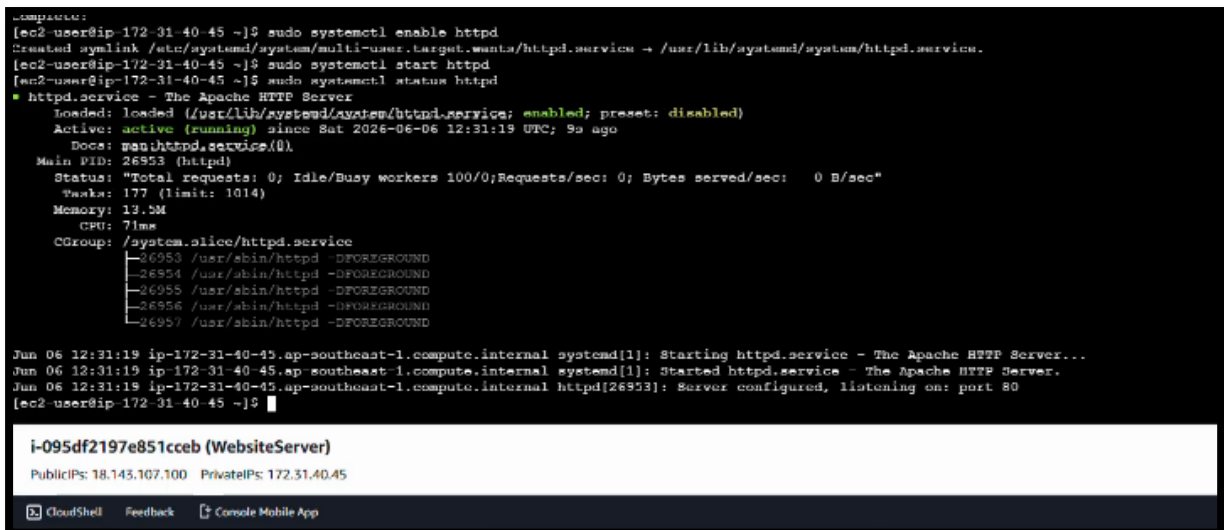
SSH was used to establish a secure connection between the local computer and the EC2 instance for server configuration and management.



```
Amazon Linux 2023
https://www.amazon.com/linux/amazon-linux-2023

Last login: Sat Jun 6 12:27:38 2026 from 3.0.5.36
[ec2-user@ip-172-31-40-45 ~]$ sudo dnf update -y
Amazon Linux 2023 Patrol Zsync repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-40-45 ~]$ cat /etc/os-release
NAME="Amazon Linux"
VERSION="2023"
ID="amzn"
ID_LIKE="fedora"
VERSION_ID="2023"
PLATFORM_ID="platform12023"
PRETTY_NAME="Amazon Linux 2023.11.20260226"
ANSI_COLOR="0;33"
CVE_NAME="CVE-2.3.0:amazon/amazon-linux-2023"
HOME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
DOCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
SUPPORT_URL="https://aws.amazon.com/premiumsupport/"
BUG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
VENDOR_URL="https://aws.amazon.com/"
[ec2-user@ip-172-31-40-45 ~]$ sudo dnf install httpd -y
Last metadata expiration check: 0:00:40 ago on Sat Jun 6 12:30:14 2026.
```

Apache web server was installed on the EC2 instance to enable website hosting.



```
[ec2-user@ip-172-31-40-45 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-40-45 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-40-45 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-06-06 12:31:19 UTC; 9s ago
     Docs: man:httpd.service(8).
  Main PID: 26953 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1014)
    Memory: 13.5M
       CPU: 71ms
    CGroup: /system.slice/httpd.service
            └─26953 /usr/sbin/httpd -DFOREGROUND
              26954 /usr/sbin/httpd -DFOREGROUND
              26955 /usr/sbin/httpd -DFOREGROUND
              26956 /usr/sbin/httpd -DFOREGROUND
              26957 /usr/sbin/httpd -DFOREGROUND

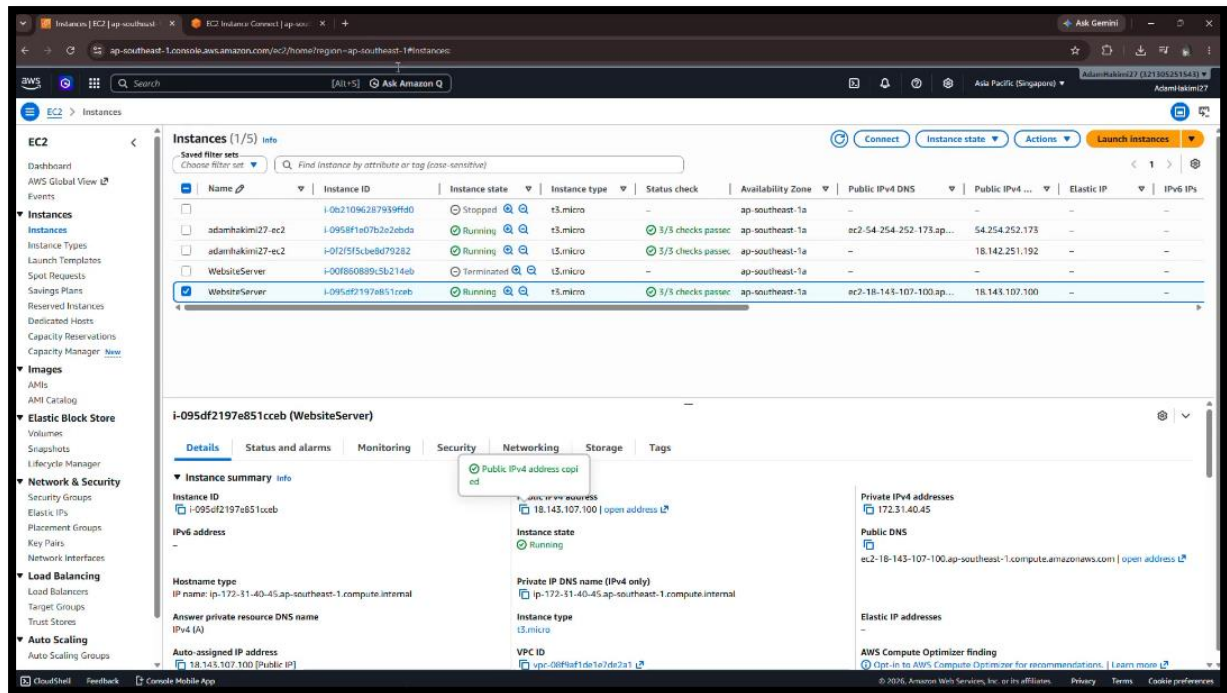
Jun 06 12:31:19 ip-172-31-40-45.ap-southeast-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jun 06 12:31:19 ip-172-31-40-45.ap-southeast-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jun 06 12:31:19 ip-172-31-40-45.ap-southeast-1.compute.internal httpd[26953]: Server configured, listening on: port 80
[ec2-user@ip-172-31-40-45 ~]$
```

i-095df2197e851cceb (WebsiteServer)

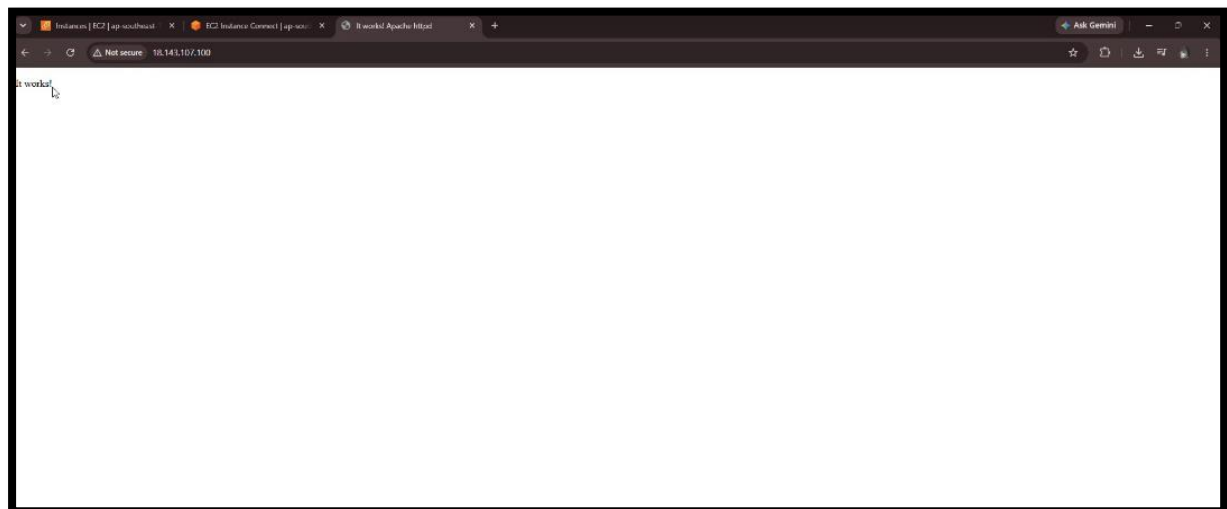
Public IPs: 18.143.107.100 Private IPs: 172.31.40.45

CloudShell Feedback Console Mobile App

Apache was successfully enabled and started on the EC2 instance, and the service status confirmed that the web server was running correctly.



The EC2 instance was successfully deployed and assigned a public address for internet access.



The Apache web server was successfully configured on the EC2 instance. Accessing the server's public IP address displayed the default "It works!" page, confirming that the web server was operating correctly.

```
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.67-3.amzn2023.0.1.x86_64 6/13
Verifying : httpd-core-2.4.67-3.amzn2023.0.1.x86_64 7/13
Verifying : httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch 8/13
Verifying : httpd-tools-2.4.67-1.amzn2023.0.1.x86_64 9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_httpd-2.0.39-1.amzn2023.0.1.x86_64 12/13
Verifying : mod_lua-2.4.67-1.amzn2023.0.1.x86_64 13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.3-1.amzn2023.0.2.x86_64          apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64          apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.67-1.amzn2023.0.1.x86_64          httpd-core-2.4.67-1.amzn2023.0.1.x86_64          httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch
httpd-tools-2.4.67-1.amzn2023.0.1.x86_64          libbrotli-1.0.9-4.amzn2023.0.2.x86_64          mailcap-2.1.49-3.amzn2023.0.3.noarch          mod_httpd-2.0.39-1.amzn2023.0.1.x86_64
mod_lua-2.4.67-1.amzn2023.0.1.x86_64

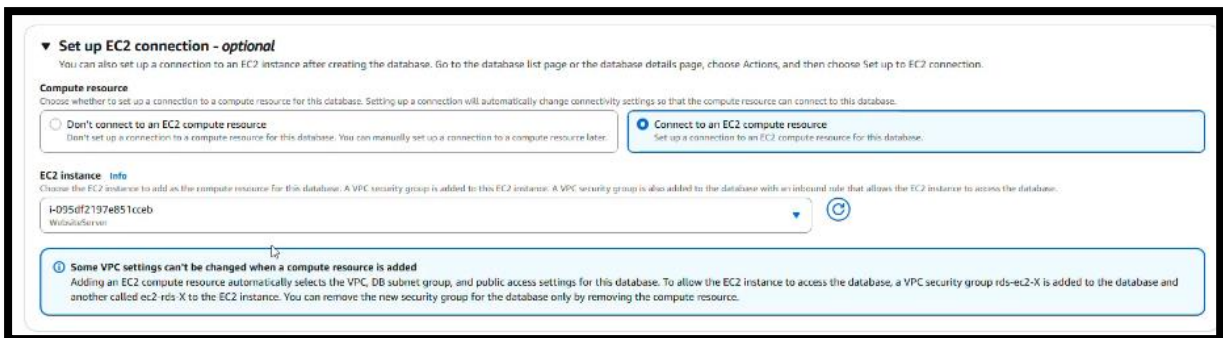
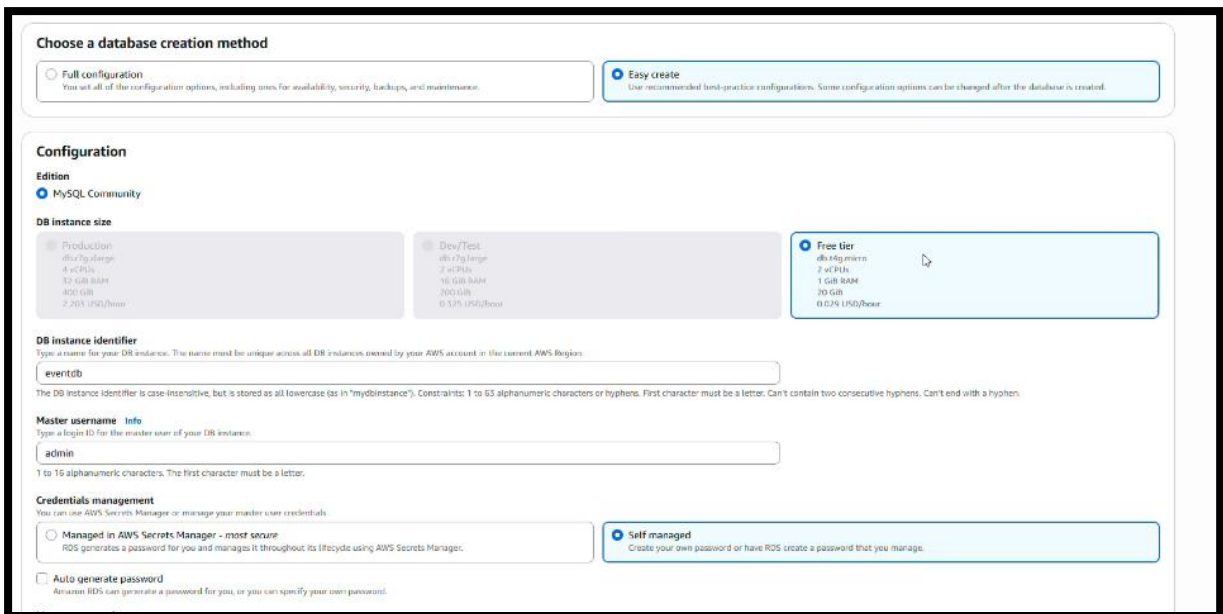
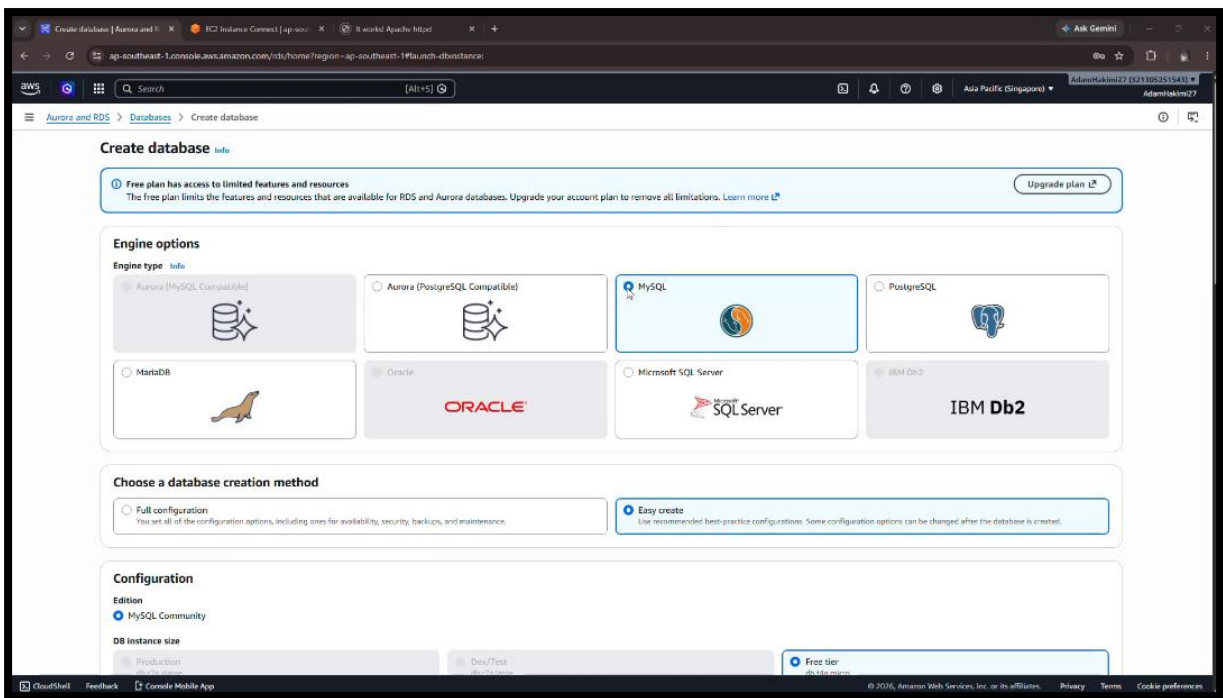
Complete!
[ec2-user@ip-172-31-40-45 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-40-45 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-40-45 ~]$ sudo systemctl status httpd
* httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-06-06 12:31:19 UTC; 9s ago
     Docs: man:httpd.conf(8)
           https://httpd.apache.org/docs/2.4/
   Main PID: 24952 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1014)
    Memory: 12.5M
       CPU: 7ms
   CGroup: /system.slice/httpd.service
           └─ 24953 /usr/sbin/httpd -DFOREGROUND
             24954 /usr/sbin/httpd -DFOREGROUND
             24955 /usr/sbin/httpd -DFOREGROUND
             24956 /usr/sbin/httpd -DFOREGROUND
             24957 /usr/sbin/httpd -DFOREGROUND

Jun 06 12:31:19 ip-172-31-40-45.ap-southeast-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jun 06 12:31:19 ip-172-31-40-45.ap-southeast-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server...
Jun 06 12:31:19 ip-172-31-40-45.ap-southeast-1.compute.internal httpd[24953]: Server configured, listening on: port 80
[ec2-user@ip-172-31-40-45 ~]$ sudo apt install php libapache2-mod-php php-mysql mysql-client -y
sudo apt: command not found
[ec2-user@ip-172-31-40-45 ~]$ sudo apt install php libapache2-mod-php php-mysql mysql-client -y
sudo apt: command not found
[ec2-user@ip-172-31-40-45 ~]$ sudo dnf install php php-mysql -y
Last metadata expiration check: 0:00:05 ago on Sat Jun 6 12:30:14 2026.
i
i-095df2197e851cceb (WebsiteServer)
PublicIPs: 18.143.107.100 PrivateIPs: 172.31.40.45
CloudShell Feedback Console Mobile App
© 2025 Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
```

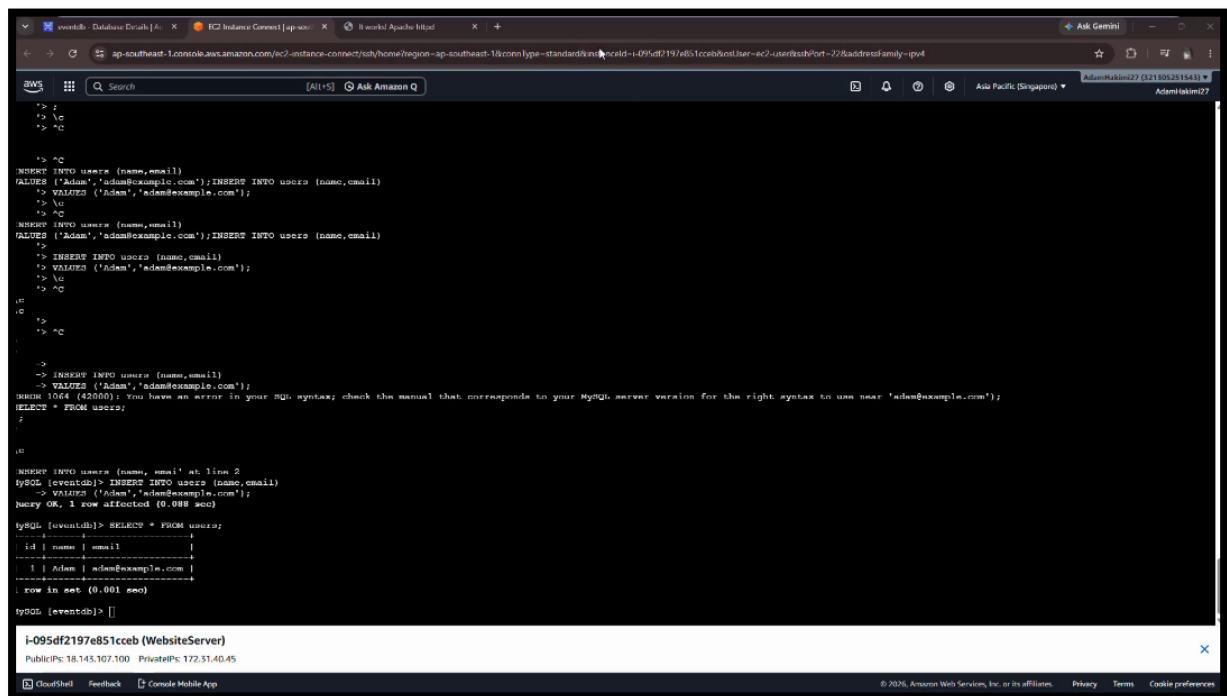
Installation of PHP and MySQL packages on the EC2 instance.

```
Complete!
[ec2-user@ip-172-31-40-45 ~]$ sudo dnf install mariadb105 -y
Last metadata expiration check: 0:00:27 ago on Sat Jun 6 12:30:14 2026.
i
i-095df2197e851cceb (WebsiteServer)
PublicIPs: 18.143.107.100 PrivateIPs: 172.31.40.45
CloudShell Feedback Console Mobile App
```

Installation of the MariaDB client on the EC2 instance.



Create Database



The screenshot shows an AWS CloudShell terminal window with the following commands and output:

```
> ?
> \c
> ^C

> ^C
INSERT INTO users (name,email)
VALUES ('Adam','adam@example.com');INSERT INTO users (name,email)
> VALUES ('Adam','adam@example.com');
> \c
> ^C
INSERT INTO users (name,email)
VALUES ('Adam','adam@example.com');INSERT INTO users (name,email)
>
> INSERT INTO users (name,email)
> VALUES ('Adam','adam@example.com');
> \c
> ^C

mysql [eventdb]>
> ^C

>
> INSERT INTO users (name,email)
> VALUES ('Adam','adam@example.com');
mysql [eventdb]> SELECT * FROM users;
SELECT * FROM users;

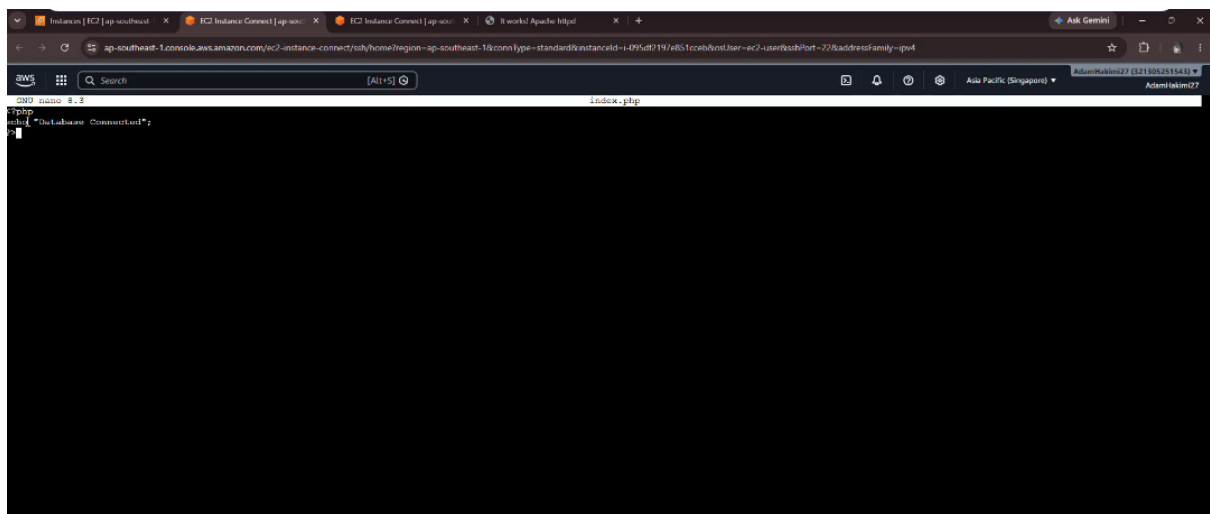
mysql [eventdb]> INSERT INTO users (name,email)
VALUES ('Adam','adam@example.com');
mysql [eventdb]> SELECT * FROM users;
+----+-----+-----+
| id | name | email |
+----+-----+-----+
| 1 | Adam | adam@example.com |
+----+-----+-----+
1 row in set (0.001 sec)

mysql [eventdb]> []
```

Below the terminal output, a banner for the EC2 instance is displayed:

```
i-095df2197e851cceb (WebsiteServer)
Public IP: 18.143.107.100 Private IP: 172.31.40.45
```

Testing Database connection



The screenshot shows an AWS CloudShell terminal window with the following commands and output:

```
> ?
> \c
> ^C

> ^C
INSERT INTO users (name,email)
VALUES ('Adam','adam@example.com');INSERT INTO users (name,email)
> VALUES ('Adam','adam@example.com');
> \c
> ^C
INSERT INTO users (name,email)
VALUES ('Adam','adam@example.com');INSERT INTO users (name,email)
>
> INSERT INTO users (name,email)
> VALUES ('Adam','adam@example.com');
> \c
> ^C

mysql [eventdb]>
> ^C

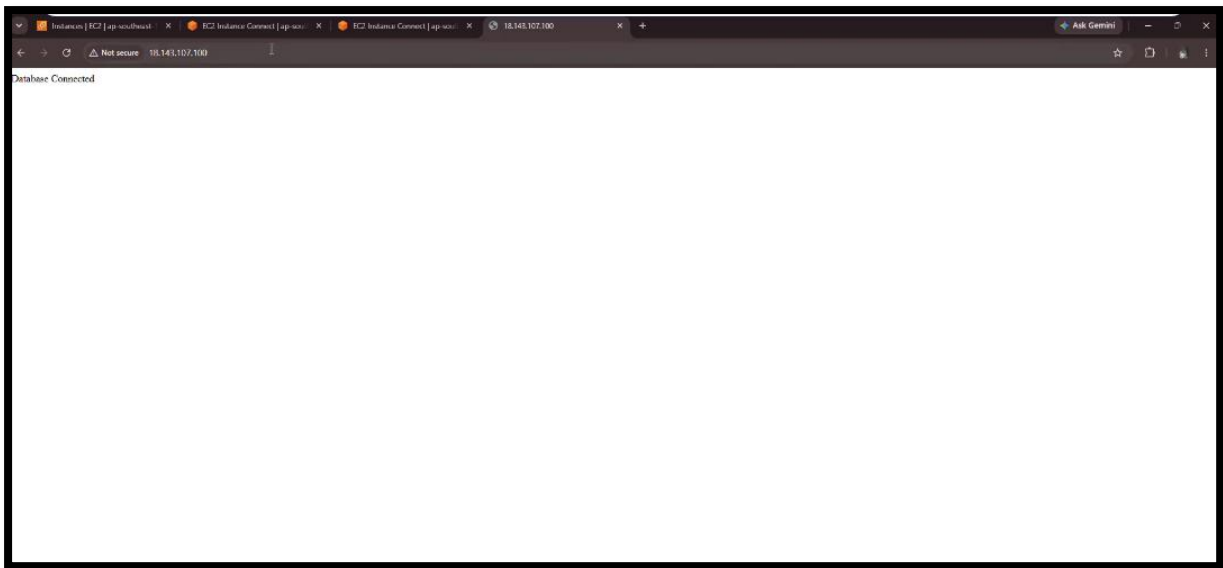
>
> INSERT INTO users (name,email)
> VALUES ('Adam','adam@example.com');
mysql [eventdb]> SELECT * FROM users;
SELECT * FROM users;

mysql [eventdb]> INSERT INTO users (name,email)
VALUES ('Adam','adam@example.com');
mysql [eventdb]> SELECT * FROM users;
+----+-----+-----+
| id | name | email |
+----+-----+-----+
| 1 | Adam | adam@example.com |
+----+-----+-----+
1 row in set (0.001 sec)

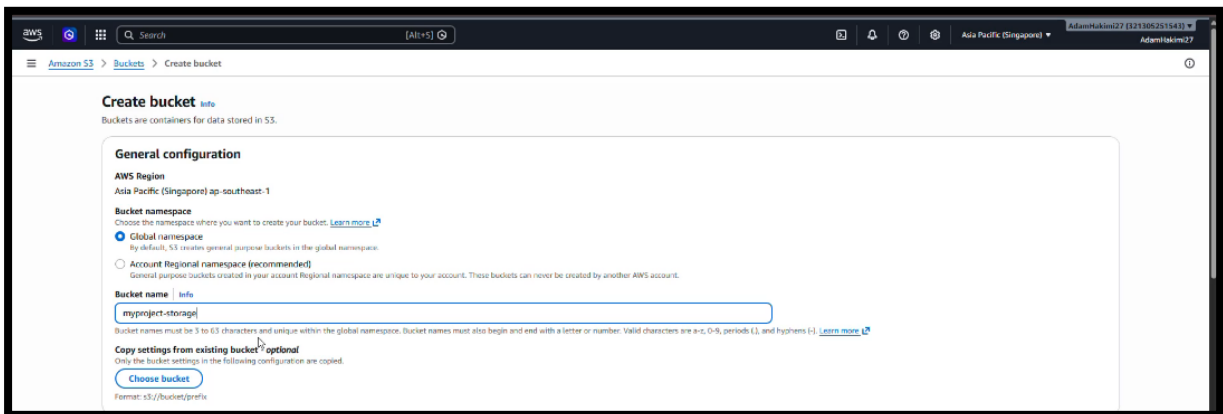
mysql [eventdb]> []
```

Below the terminal output, a banner for the EC2 instance is displayed:

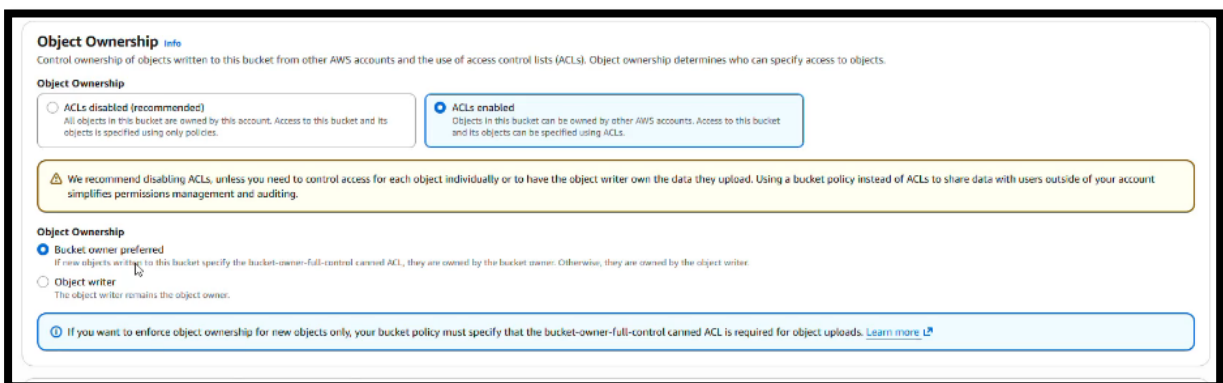
```
i-095df2197e851cceb (WebsiteServer)
Public IP: 18.143.107.100 Private IP: 172.31.40.45
```



Using the same public IP and test the connection to see the database connection.



Creating Bucket



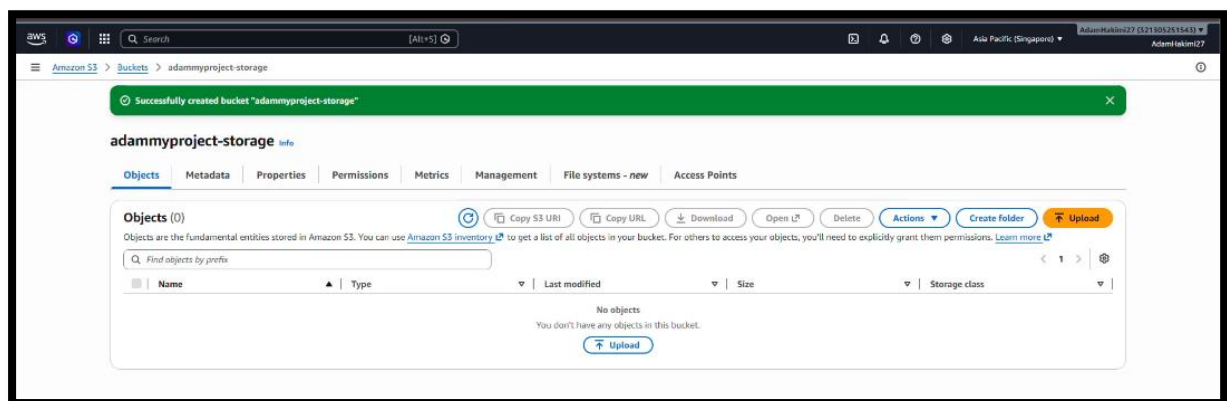
Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

- ☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
 - ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
 - ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
 - ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
 - ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

⚠ Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

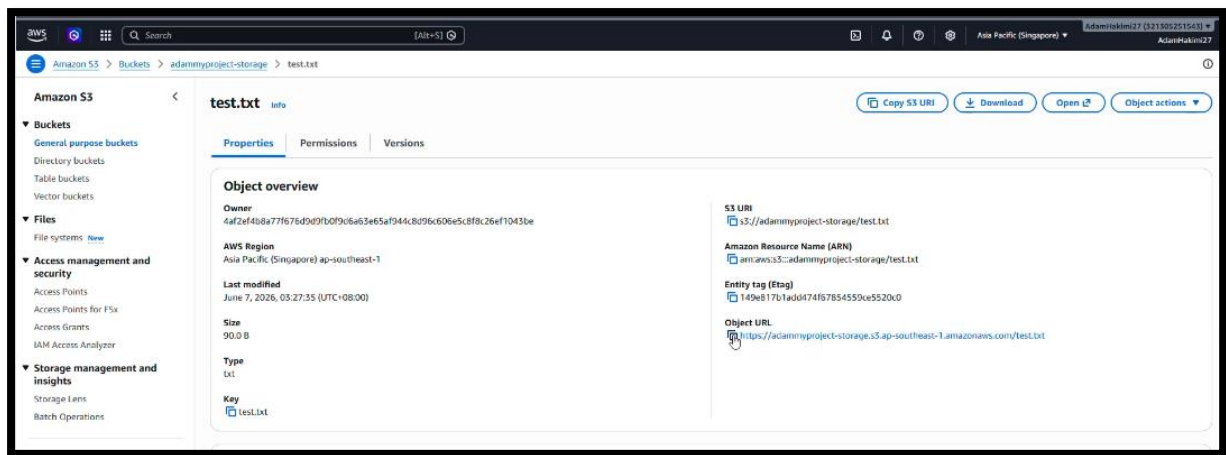
☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.



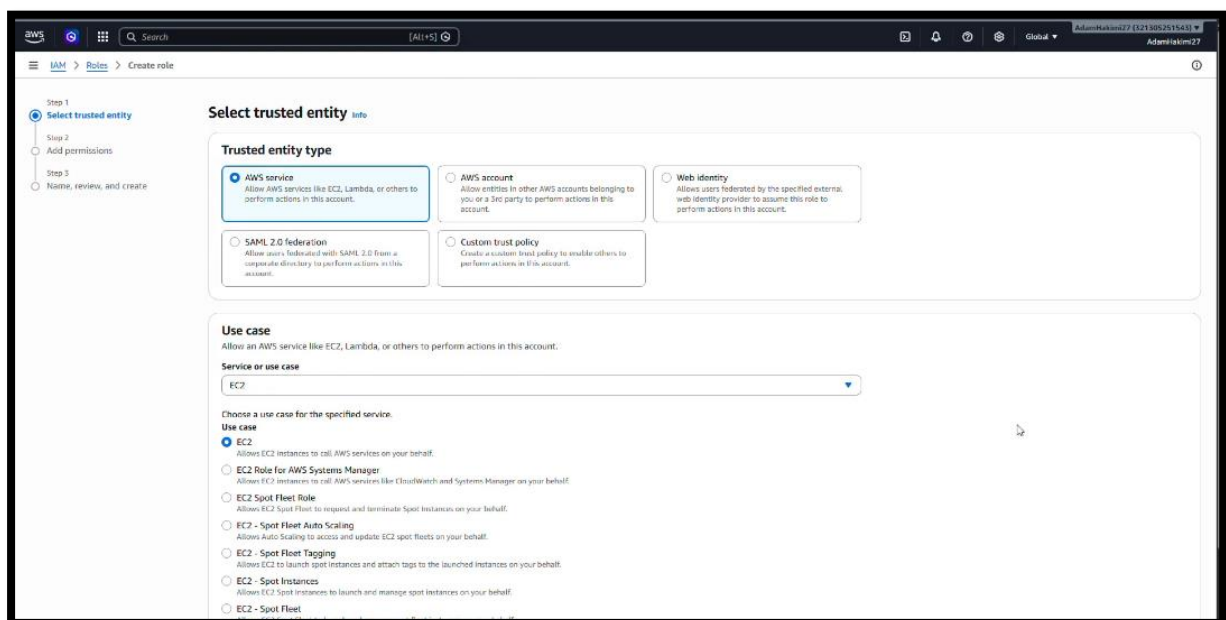
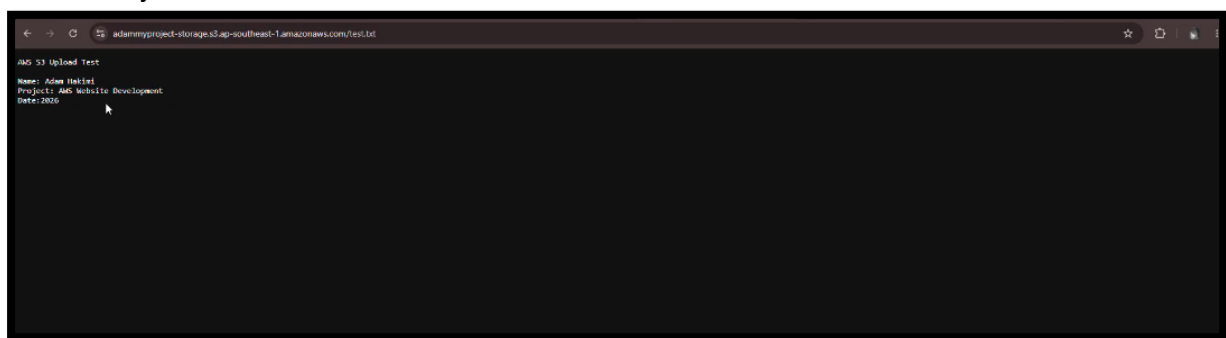
```

File Edit View H1 1 To do AWS S
AWS S3 Upload Test
Name: Adam Hakimi
Project: AWS Website Development
Date: 2026
  
```

Creating txt file. For S3 upload test. Upload this txt file in the object.



Use the object URL.



Creating IAM role.

```

[ec2-user@ip-172-31-40-45 ~]$ aws s3 ls
2026-06-06 19:31:03 adammyproject-storage
2026-05-25 01:54:20 my-test-bucket-adambakimi27
[ec2-user@ip-172-31-40-45 ~]$ echo "Hello from EC2" > test-ec2.txt
[ec2-user@ip-172-31-40-45 ~]$ cat test-ec2.txt
Hello from EC2
[ec2-user@ip-172-31-40-45 ~]$ aws s3 cp test-ec2.txt s3://adammyproject-storage/
upload: ./test-ec2.txt to s3://adammyproject-storage/test-ec2.txt
[ec2-user@ip-172-31-40-45 ~]$ aws s3 ls s3://adammyproject-storage/
2026-06-06 19:40:46      15 test-ec2.txt
2026-06-06 19:27:35      90 test.txt
[ec2-user@ip-172-31-40-45 ~]$

```

A test file was successfully uploaded from the EC2 instance to the Amazon S3 bucket, confirming that EC2 could access and interact with S3.

```

[ec2-user@ip-172-31-40-45 ~]$ mysql -h eventdb.cwawqy0t40e.ap-southeast-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \q.
Your MySQL connection id is 84
Server version: 8.0.42 Source Distribution

Copyright (c) 2000, 2019, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\q' to clear the current input statement.

MySQL [(none)]> USE eventdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [(none)]> SELECT * FROM users;
+----+-----+-----+
| id | name | email |
+----+-----+-----+
| 1  | Adam | adam@example.com |
+----+-----+-----+
1 row in set (0.002 sec)

MySQL [(eventdb)]>

```

i-095df2197e851cceb (WebsiteServer)
PublicIP: 18.143.107.100 PrivateIP: 172.31.40.45

The EC2 instance successfully connected to the RDS database and data was retrieved from the users table, confirming successful database integration.

```

ec2-user@ip-172-31-40-45 ~]$ aws s3 ls s3://adammyproject-storage/
2026-06-06 19:40:46      15 test-ec2.txt
2026-06-06 19:27:35      90 test.txt
ec2-user@ip-172-31-40-45 ~]$ sudo systemctl status httpd
Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
Drop-In: /usr/lib/ systemd/systemd/httpd.service.d
         .drop-in: /usr/lib/ systemd/systemd/httpd.service.d
Active: active (running) since Sat 2026-06-06 19:18:02 UTC; 28min ago
Docs: man:httpd.service(8)
Main PID: 47325 (httpd)
Status: "Total requests: 8; Idle/Busy workers 100/0;Requests/sec: 0.00473; Bytes served/sec: 2 B/sec"
Tasks: 230 (limit: 1014)
Memory: 17.4M
CGroup: /system.slice/httpd.service
┌─47325 /usr/sbin/httpd -DFOREGROUND
┌─47328 /usr/sbin/httpd -DFOREGROUND
┌─47329 /usr/sbin/httpd -DFOREGROUND
┌─47329 /usr/sbin/httpd -DFOREGROUND
┌─47329 /usr/sbin/httpd -DFOREGROUND
└─47308 /usr/sbin/httpd -DFOREGROUND

Jun 06 19:18:01 ip-172-31-40-45.ap-southeast-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jun 06 19:18:02 ip-172-31-40-45.ap-southeast-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jun 06 19:18:02 ip-172-31-40-45.ap-southeast-1.compute.internal httpd(47325): Server configured, listening on port 80
ec2-user@ip-172-31-40-45 ~]$

```

i-095df2197e851cceb (WebsiteServer)
PublicIP: 18.143.107.100 PrivateIP: 172.31.40.45

The Apache web server was verified to be running successfully on the EC2 instance.

Summary

This project successfully demonstrated the deployment of a cloud-based web application using Amazon Web Services (AWS). The infrastructure was built using Amazon EC2 as the web server, Amazon RDS MySQL as the database service, and Amazon S3 as cloud storage. A security group was configured to allow SSH, HTTP, and HTTPS traffic for secure server management and web access.

The Apache web server, PHP, and MariaDB client were installed and configured on the EC2 instance. The website successfully hosted and verified through the EC2 public IP address. An Amazon RDS MySQL database was created, connected to the EC2 instance and tested by creating tables and inserting sample data.

Amazon S3 was integrated into the project to provide cloud storage functionality. Files were uploaded both manually and through the AWS CLI from the EC2 instance, confirming successful communication between EC2 and S3. Finally, all components were tested, including website accessibility, database connectivity, Apache service status, and S3 operations. The successful completion of these tests demonstrated that the cloud infrastructure was fully functional and met project objectives.